

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Multivariable Calculus	Course Code:	MT2008
	Program:	Electrical Engineering	Semester:	Fall 2023
	Duration:	3 hour	Total Marks:	70
	Exam Date:	December 30 th , 2023	Weight:	50
	Section:	All	Page(s):	
	Exam Type:	Final	CLO #	1, 2, 3

Student Name: _____ Roll No: _____ Section: _____
 Instruction/Notes: 1. Do not forget to write your Name and Roll Number.

Question 1: (CLO 01) (marks: 5+5+5)

- Formulate the parametric equations and symmetric equations for the line from $(1, -1, 1)$ and parallel to the line $x + 2 = \frac{1}{2}y = z - 3$.
- Find an equation of the plane that passes through the point $(1, -1, 1)$ and contains the line with symmetric equations $x = 2y = 3z$.
- Calculate the distance between the given parallel planes $6z = 4y - 2x$, $9z = 1 - 3x + 6y$.

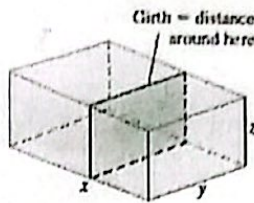
Question 2: (CLO 02) (marks: 5+10+10)

- If the resistors of R_1, R_2 , and R_3 ohms are connected in parallel to make an R -ohm resistor, the value of R can be found from the equation

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Find the value of $\partial R / \partial R_2$ when $R_1 = 30, R_2 = 45$, and $R_3 = 90$ ohms.

- Suppose that the temperature at a point (x, y, z) in space is given by $T(x, y, z) = 200e^{-x^2-3y^2-9z^2}$ where T is measured in degrees Celsius and x, y, z in meters. In which direction does the temperature increase fastest at the point $(1, 1, -2)$? What is the maximum rate of increase?
- A delivery company accepts only rectangular boxes the sum of whose length and girth (perimeter of a cross-section) does not exceed 108 inches. Find the dimensions of an acceptable box of largest volume.



Question 3: (CLO 03) (marks: 10+10+10)

- a) Find area of the part of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 1$.
- b) Use Green's Theorem to calculate the work done by force field in moving a particle over a triangle from $(0, 0)$ to $(0, 4)$ to $(2, 0)$ to $(0, 0)$ where $F = (y \cos x - xy \sin x, xy + x \cos x)$. (Check the orientation of the curve before applying the theorem.)
- c) Calculate the flux of vector field $F(x, y, z) = 3xy^2\mathbf{i} + xe^z\mathbf{j} + z^3\mathbf{k}$ across the surface of the solid bounded by the cylinder $y^2 + z^2 = 1$ and planes $x = -1$ and $x = 2$.